BEN HE

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Profile

M.Sc. student in Electrical Engineering & Information Technology at KIT, focused on robotics software and applied LLM systems. Experience ranging from microcontrollers and CAN bus to ROS 2, digital twins and LLM agents in practical use. Looking for a working-student role, internship or thesis project in this area.

Education

Karlsruhe Institute of Technology (KIT) **10/2026 – present**
M.Sc. Electrical Engineering & Information Technology (EEIT) *Karlsruhe*

Karlsruhe Institute of Technology (KIT) **10/2021 – 06/2026**
B.Sc. Electrical Engineering and Information Technology · Thesis grade: 2.3 (German scale) *Karlsruhe*

Study Preparation in Germany **10/2020 – 09/2021**
telc Deutsch C1 Hochschule (04/2021)

Yangtze University **09/2017 – 09/2020**
Studies in Information and Computing Science (transferred to KIT) *Jingzhou, China*

Professional Experience

Warehouse Management System Developer **07/2026 – 10/2026**
Buddhiherz / König Winna GmbH (Ettlingen) *Python, Flask, SQLite, LLM, nginx, GLS API*

- Three-stage document extraction: offline rules (≈0.02 s, no API cost), then LLM text and image extraction with a fallback model; edge cases go to a manual correction workflow.
- Delivered with Flask, SQLite, Waitress, systemd and nginx on a Linux VPS; integrated the GLS API for shipping labels and built PDF generation for invoices and delivery notes.
- Replaces the manual Excel and paper process in the warehouse; role-based login (staff via email + PIN), daily backups and 100+ automated regression tests.

R&D Intern – Embedded Systems **12/2025 – 02/2026**
Dynavin GmbH (Karlsruhe) *MCU, CAN bus, debugging*

- Analysis of electronic circuits for vehicle aftermarket systems: 32-bit microcontroller, CAN-bus interface, power supply and input/output stages.
- Hardware assembly, systematic troubleshooting and repair of control units in the test workshop; functional testing of navigation systems.
- Support for vehicle integration of aftermarket systems and diagnosis of CAN-bus components in customer vehicles.

University Projects

Bachelor's Thesis: Design and Implementation of an AAS Submodel for Industrial Robotic Systems **06/2025 – 12/2025**
Institute of Control Systems (IRS), KIT · Grade 2.3 · *Python, ROS 2, AAS v3.0, URDF/Xacro*
github.com/BenHe2019/URCell

- Python toolchain that automatically converts URDF/YAML data into six semantic AAS submodels (Structure, Kinematics, Dynamics, Control, Safety, Visualization) for 12 UR models and the igus ReBeL.
- Multi-robot generator that automatically builds ROS 2 simulation scenes from a configuration file (type, position, joint limits) and the AAS data.
- End-to-end validation in simulation: Universal Robots (UR3/UR5) via the standard generator, igus ReBeL via an extended pipeline.

Hardware Design & Power Electronics **05/2024 – 09/2024**

Institute of Electrical Engineering (ETI), KIT *MATLAB/Simulink, Altium Designer, STM32, Embedded Coder*

- Buck converter from design to commissioning: 40 V input, 1–36 V output, up to 50 W; simulated in MATLAB/Simulink (Simscape), design target for output ripple ≤ 0.5 % of V_{in}.
- Multi-layer PCB in Altium Designer (schematic, layout, DRC), hand assembly and systematic commissioning.
- Cascaded PI control (current and voltage) with Stateflow fault shutdown on an STM32; C++ code generated with Embedded Coder.

Embedded Model Car: Distance Sensing & Steering **05/2023 – 07/2023**

Institute for Information Processing Technology (ITIV), KIT *C++, embedded systems, Tiva C*

- Team project in the Information Technology I lab: my own C++ modules for ultrasonic distance sensing (HC-SR04, timer capture) with multi-level LED warning and for joystick steering (ADC, differential drive) on a Tiva C microcontroller (TM4C123).

Personal Projects

Bennu — Self-Hosted LLM Voice Assistant (Robot Brain) **09/2026 – present**

In development *Python, LLM agents, STT/TTS, Android, smart home*

- Voice chain (wake word → ASR → LLM agent → TTS) with a fast path for common commands, plus calendar, bookkeeping, health data and smart-home control; voice, web dashboard, Android app and sensors all feed into one database.
- Long-term goal: the brain of a home robot; the robot body is still in planning.

Fitlog — Android App for Nutrition and Training **01/2025 – 06/2026**

Private repository · access on request *Kotlin/Java, MVVM, Room, Python, multimodal LLMs*

- Android app with a Room database, REST API and offline-capable sync; estimates calories from food photos via a multimodal LLM, with results reviewed by the user before saving.

Skills

AI & LLM	LLM agents, multimodal LLMs, prompt engineering
Robotics	ROS 2, URDF/Xacro, AAS / digital twin
Programming	Python, C++, Kotlin/Java, MATLAB
Embedded & Hardware	STM32, CAN bus, Altium Designer, MATLAB/Simulink
Mobile & Web	Android SDK, MVVM, Room, REST API design
Tools	Git, Docker, Linux (Ubuntu), LaTeX
Languages	Chinese (native), German (C1, telc C1 Hochschule), English (B2)